

Certified but Compromised: Breaking and Fixing Lightweight Proofs of Inference

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Abstract

A company that outsources inference to a cloud provider cannot tell whether the provider ran the agreed model or changed the answer on purpose. Proofs of inference let it check each answer. Those built on zkSNARKs give strong guarantees but take minutes per query on large models, motivating lighter protocols that check only a sample of the computation, such as that of Anchuri et al., where the client re-checks one random path through a committed execution trace. We show that sampling misses targeted changes: a provider that tampers with a single neuron and computes everything else honestly almost always passes the path check, on image classifiers and on a real large language model. No sampling rule, however it chooses where to look, closes this gap on every network. We therefore build a protocol that checks every layer with Freivalds’ algorithm against a Reed–Solomon/Merkle commitment to the weights. It needs no SNARK, trusted hardware or trusted setup, a wrong output passes with negligible probability, and in our experiments it rejected thousands of attacks without a miss. Its prover is orders of magnitude faster than published zkSNARK provers and faster than every other system we compare with, and its verifier is faster than most of them and competitive with the rest; its main cost is a proof that grows with the input.

Keywords

verifiable inference, proof of inference, integrity attacks, Freivalds’ algorithm, zkML

1 Introduction

Consider a company that pays a cloud provider to serve a licensed or fine-tuned language model to its users. Nothing in a response shows that the provider ran that model with the agreed weights. The provider may cut costs by serving a cheaper quantised copy, but it may also return a wrong answer on purpose. A company that handles sensitive data and acts on the model’s answers, for example to approve payments, flag fraud or triage medical records, is exposed to a provider, or an attacker inside it, that changes the answers to chosen queries and serves every other query correctly. The company cannot re-run the model to find out, since it lacks the weights and the hardware; that is why it outsourced inference. A *proof of inference* closes this gap. With each answer the provider sends a proof, and the client (here, the company) runs a check, much cheaper than inference, that fails with high probability unless the answer was computed with the agreed weights on this input.

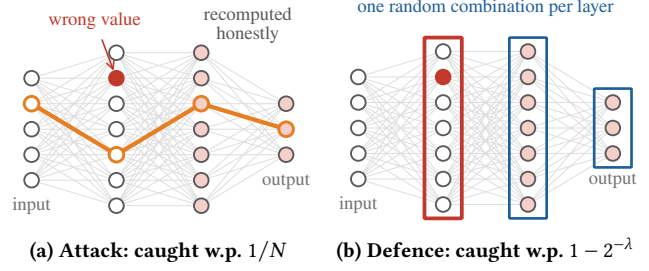


Figure 1: One tampered neuron slips past a random path, but not past a check of every layer

zkSNARKs [8, 15, 25, 29, 31] prove that the answer is correct and can keep the weights secret. Their proofs are usually small and fast to check, but producing one takes far longer than inference; zkLLM needs about ten minutes on an A100 for one Llama-2-7B prompt [31]. Anchuri et al. [2] proposed a much lighter approach. The provider commits to the full execution trace with a Merkle tree, and the client checks the computation along *one random path* from an output neuron back to the input, against a commitment to the weights. Proving drops from minutes to milliseconds. A different model produces a trace that differs from the correct one in many places, and a random path is likely to hit one of them. Anchuri et al. accordingly prove *other-model* soundness, against a provider that honestly runs a different model. They note that one path cannot catch a single changed node with probability above $1/N$ in a layer of width N , and they leave *full* soundness, against any wrong answer, open for LLMs.

We present an attack on this protocol, which we implement and evaluate. A provider that keeps the committed model and changes only what it needs to can overwrite *a single neuron* and recompute everything after it honestly (Figure 1a). The trace is then inconsistent at one node, which a random path visits with probability $1/N$, so almost every tampered answer is accepted. Applied only to inputs that carry a trigger, the edit is a backdoor in the serving code, which leaves the committed weights clean and answers clean queries bit for bit as before. The attack carries over to the real OPT-6.7B. Checking many queries does not close the gap, because the provider chooses which queries to tamper with, and tampering with k of them is caught with probability at most k/N . Nor do the fixes Anchuri et al. suggest, sampling more paths or sampling where it matters: no rule does better than uniform sampling on every network, and reaching 2^{-40} forces the path protocol to open essentially the whole model and trace.

This led us to a different design (Figure 1b). The client tests one random linear combination of all the neurons of every layer (Freivalds’ algorithm [14]). When it does not hold the weights, it checks the combination against a Reed–Solomon-encoded, Merkle-committed copy of them in the style of Ligerio [1], so, as in the path protocol, it stores 32 bytes per weight matrix. The model runs as an int8 network whose results are bit-identical on the provider’s GPU and on the client, which makes the test exact. A tampered value anywhere in the network is then caught with probability $1 - 2^{-\lambda}$ instead of $1/N$, and the provider’s extra work grows with the model but not with the input. At 2^{-128} our proofs are also smaller than the paths’ at 2^{-40} : catching the single-neuron attack on a 64-token Llama-2-7B prompt with probability $1 - 2^{-40}$ takes paths that open about 13.7 GB, while our proof is 325 MB.

Contributions. (1) **An attack** (Sections 3.2 and 4.2) that exploits the $1/N$ limit acknowledged in [2]. One tampered neuron is accepted with probability $1 - 1/N$ (99.6–99.8% on MNIST) and changes the next token of 40 of 40 prompts on the real OPT-6.7B. (2) **A bound on sampling-based fixes** (Section 3.3): no sampler beats uniform sampling on every network, and uniform sampling needs essentially the whole model and trace for 2^{-40} . (3) **A full-soundness protocol** for whole CNNs and transformers (Sections 3.4–3.6), with error $2^{-\lambda}$, that composes known checks and needs only a hash commitment per weight matrix (Section 2). Optimisations keep the bound and shrink the committed-weights proof 2.0–4.2× for images and prompts of up to 512 tokens. (4) **A matched-model evaluation** (Section 4) on six image classifiers and 10 LLM architectures up to 13B parameters, against [2], 17 published systems and zkLLM’s code on our GPU. Our prover is 35–32,000× faster than the zkSNARK provers. Code and raw measurements are released.

2 Related Work

zkSNARK proofs of inference. A long line of work compiles neural networks into zero-knowledge proofs. For CNNs these include sum-check/GKR provers [17] such as zkCNN [25] and zkPyTorch [35], and pairing- or Plonk-based systems such as vCNN [23], ZEN [11], ZENO [12], ZKML [8], EZKL [39] and Bionetta [38]; Mystique [34] is an interactive, VOLE-based alternative. For LLMs they include zkLLM [31] (also interactive), zkGPT [29], DeepProve [15], ZKTorch [7] and Jolt Atlas [4]. They give computational soundness and mostly small proofs, but their provers are far slower than inference: from half a second for LeNet-5 to minutes or hours for larger models.

Freivalds-style checks. SafetyNets [16] uses interactive proofs for networks with quadratic activations. Slalom [33] checks every linear layer of a GPU computation from inside an SGX enclave with Freivalds’ algorithm, optionally with a precomputed secret challenge; TwinShield [36] and VeriAttn [9] extend this to attention inside TEEs. Several recent works need no enclave. Maverick [26] checks every linear layer of a language model with sparse Freivalds checks on a code-encoded copy of the weights that the client stores, drawing a fresh challenge per query. Gupta et al. [19] run Freivalds’ check in the exponent of a group, against verification data that the model owner generates with secret randomness, for a client that outsources inference to two non-colluding servers.

CommitLLM [6] checks int8 language models with secret Freivalds vectors precomputed from the weights, audits sampled positions and layers, and leaves attention unverified. LAMP [22] lets a verifier that holds only a commitment check one matrix product, using Freivalds’ check and Reed–Solomon proximity testing inside a Groth16 SNARK.

Sampling, fingerprints and optimistic protocols. Anchuri et al. [2] spot-check random paths and argue security through trace separation; SLP [24] samples which layers receive a SNARK. TOPLOC [28] and SVIP [32] fingerprint hidden states, with empirical guarantees. Verde [3], opML [10] and TAO [37] resolve disputes and assume an honest party or an active challenger.

Where we fit. We keep the setting of [2], a client that holds a 32-byte commitment per weight matrix, but replace its path test, which we break (Sections 3.2 and 3.3), with full soundness $2^{-\lambda}$ for complete CNNs and transformers (Sections 3.4 and 3.5). We chain known protocols through every layer with an end-to-end soundness proof: Freivalds’ check, and the column check of the Ligerio and Brakedown commitments [1, 18]. The protocols above that check every layer without a SNARK each need more than a hash commitment: Slalom an enclave, Maverick an encoded copy of the weights, CommitLLM a secret key derived from them, and Gupta et al. public data that grows with every matrix and is generated with secret randomness. zkSNARKs that verify against a Merkle root, such as DeepProve’s transparent configuration, need no trusted setup but prove far more slowly. To our knowledge, ours is the first protocol that is fully sound for complete CNNs and transformers using only a hash function and a linear code, with no SNARK, TEE or trusted setup.

3 Technical Approach

3.1 Setting and threat model

A model M has a public architecture and secret or public weights. The provider acts as the prover and the client as the verifier. Before any query the client obtains a commitment C_M to the weights, for example from the model owner. As in [2], we assume that C_M is computed honestly; unlike a SNARK’s trusted setup it involves no secret randomness, and anyone who holds the weights can recompute it. For each query x the provider returns a claimed output y and a proof, and may deviate from the protocol arbitrarily. We require *completeness* (honest answers are always accepted) and *full soundness*: any $y \neq M(x)$ is rejected except with probability ϵ (Anchuri et al. prove only other-model soundness). We consider three trust settings. In setting C the client holds only C_M (32 bytes per weight matrix), as in [2] and the zkSNARKs; in K it knows the weights; and K_{pre} is K with a secret challenge precomputed once per model, as in Slalom’s preprocessing mode and CommitLLM [6]. The architecture includes the integer constants of the operations without weights (requantisation multipliers, look-up tables), which the client obtains with C_M .

3.2 The path test and the single-neuron attack

In the protocol of [2] (RandPathTest) the prover commits to every activation a_j of the forward pass with a Merkle tree. The verifier chooses a random output neuron and walks back to the input, moving to a random parent at each layer. At every node j on the path

it opens the parents’ activations and the weights w_{ij} from C_M and checks $a_j = \phi(\sum_{i \in G_j} w_{ij} a_i)$, where G_j is the set of parents of j and ϕ the activation. A path opens one weight row per layer, so the proof and the prover’s work stay small.

The single-neuron attack. Our adversary runs the correct model on x , picks one neuron j , and replaces a_j by a value that makes the output wrong (untargeted) or equal to a class of its choice (targeted). It then recomputes all later layers *with the true weights*. Every later node is consistent with its parents and every earlier node is untouched, so j is the only node whose local check fails, the only *inconsistent node*. The verifier notices the attack only if its path visits j , which happens with probability $1/N$ in a layer of width N , or $1 - (1 - 1/N)^s$ with s paths. Applied only to inputs that contain a trigger, the attack becomes a backdoor in the serving code. The prover keeps the committed weights and answers clean queries honestly, bit for bit, so no inspection of the weights reveals it and the clean-accuracy gap is zero by construction.

3.3 Why sampling does not fix it

Anchuri et al. suggest sampling more paths, or sampling where it matters instead of uniformly. Neither helps against an adversary who knows the rule. Fix a layer of width N and let U be the set of its neurons whose change can flip the output. If the sampler ignores the claimed values, the probabilities with which it opens each neuron are fixed before the adversary acts and sum to at most s over U , so the adversary can tamper with a neuron that is opened with probability at most $s/|U|$, on every network. A sampler that reads the values could do better on a particular network, but not on every one. We call a sampler *value-aware* if it chooses each neuron to open from the input, the claimed values, the weights of later layers, each neuron’s natural range (the values it takes on honest inputs) and the rows it has already opened.

PROPOSITION 3.1 (NO SAMPLER BEATS UNIFORM ON EVERY NETWORK). *For every value-aware sampler that opens at most s neurons of a hidden layer of width N and accepts every network’s honest trace, there is a network on which it catches a tampered neuron, kept within its natural range, with probability at most s/N .*

Proof. Let hidden neuron i copy input bit $\sigma(i)$ for a permutation σ , let natural inputs have at most one zero, and let the output be 1 exactly when all hidden neurons are 1. The adversary sets the neuron that copies the input’s zero to 1, which flips the output and stays in the range $\{0, 1\}$. The forged trace, later weights and ranges do not depend on σ , and the forged trace is honest for the network in which that neuron copies another bit, so the sampler rejects only if it opens the tampered row. Every other row it opens shows a neuron that copies a 1, so for a random σ the zero’s copier stays uniform among the unopened neurons and is opened with probability at most s/N ; some σ does no better. $\square$

Opening s distinct neurons uniformly reaches s/N on every dense network, so nothing beats uniform sampling in the worst case, and catching one tampered neuron with constant probability needs $s = \Omega(N)$ openings per layer, the cost the protocol was designed to avoid. The *contribution-weighted sampler*, which steps to parent i with probability proportional to $|w_{ij} a_i|$, never visits a neuron whose

claimed activation is zero, and a floor δ , sampling by $|w_{ij}|(|a_i| + \delta)$, leaves detection low (Section 4.2).

3.4 Our defence: checking every layer

Since no sampler has a better worst-case guarantee than uniform sampling, our verifier checks every neuron of every layer.

Exact integers. The checks compare exact values, but floating-point results differ between devices and summation orders. The model that is committed and checked is therefore an int8 version of M , obtained by integer-only quantisation [21], the form in which many models are already served; Section 4.1 measures its accuracy. Each weight layer ℓ computes $Z_\ell = W_\ell X_\ell + b_\ell$, where W_ℓ is int8 with one scale per output channel, X_ℓ is int8 with one scale per tensor, and BatchNorm is folded into W_ℓ and b_ℓ ; convolutions become matrix products through im2col. These integer products can still use the GPU’s fast float32 units: float32 represents every integer below 2^{24} exactly, so the prover sums at most 1,024 products of int8 values at a time, as in Slalom [33], and its results are bit-identical to the verifier’s. Exactness also removes the floating-point tolerance of the path protocol, which leaves an attacker room to hide small changes.

Three statements. With the output y , the prover sends the pre-activations Z_ℓ of every weight layer, which we call the *claims*. Write $A_\ell = [W_\ell | b_\ell]$. Between two weight layers the network applies only operations without weights: requantisation, ReLU, pooling, residual additions, LayerNorm/RMSNorm, GELU/SiLU (by table look-up), RoPE, softmax, and the attention products $QK^\top$ and PV . The answer y is the last weight layer’s pre-activations (the logits), so it is itself a claim, and it is correct if every claim is. Every claim equals the value that M computes on x if and only if three statements hold for every weight layer ℓ : (S1) its input X_ℓ follows from x and the earlier claims through these operations, for example $X_\ell = \text{requant}(\text{ReLU}(Z_{\ell-1}))$ in an MLP; (S2) its claims are the product of its weights and its input, $Z_\ell = A_\ell[X_\ell; \mathbf{1}]$; and (S3) A_ℓ is the committed matrix.

Checking (S1): recomputation. The verifier computes every operation without weights itself, from x and the claims. Each input X_ℓ is therefore the verifier’s own computation, not a value sent by the prover, and the checks are chained: a prover cannot pass one layer’s check with one value and feed another into the next. This is also why non-linear layers such as ReLU and softmax need no proof here: the verifier never checks a claimed activation, it computes it.

Checking (S2): Freivalds’ check. Let A_ℓ have N rows (one per output neuron) and row length k , and let T be the number of columns of X_ℓ (pixels or tokens); we drop the subscript ℓ from N , k and T when the layer is clear. Recomputing $A_\ell[X_\ell; \mathbf{1}]$ would cost the verifier as much as inference. Instead it draws r independent random vectors $\chi \in \mathbb{F}_p^N$ and, for each, checks one random combination of the layer’s rows,

$$\chi^\top Z_\ell = \underbrace{(\chi^\top A_\ell)}_{u_\ell} [X_\ell; \mathbf{1}], \quad (1)$$

where we call u_ℓ the *folded row*. Let $D = Z_\ell - A_\ell[X_\ell; \mathbf{1}]$ be the error of the claims: its entry D_{im} is how far the claim of neuron i at position m (a pixel or token) is from the true value, so a correct

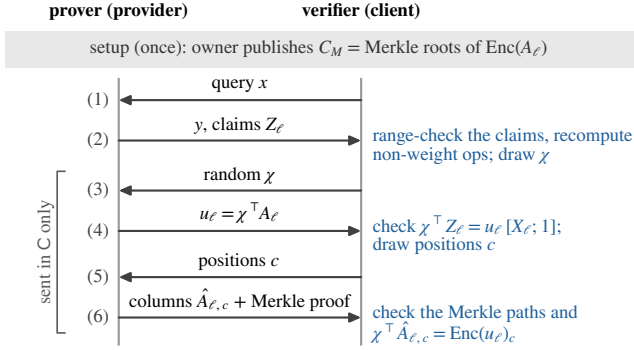


Figure 2: One query in setting C: each value is fixed before the randomness that tests it

layer has $D = 0$. The two sides of Eq. (1) differ by $\chi^\top D$, so the check passes only if $\chi^\top D = 0$. If some $D_{im} \neq 0$, entry m of $\chi^\top D$ is $\sum_{i'} \chi_{i'} D_{i'm} = \chi_i D_{im} + c$, where c does not involve χ_i . Whatever the other entries of χ are, this is zero for only one of the p values χ_i can take ($\chi_i = -c/D_{im}$), so a wrong layer passes with probability at most $1/p$; it must pass the check for each of the r vectors, so at most p^{-r} in all. This holds however many values are wrong and wherever they are: a single tampered neuron is caught as reliably as a wholly wrong layer. Given u_ℓ (cost $O(rNk)$), the two sides cost $O(r(N+k)T)$, against $O(NkT)$ for recomputing the layer.

We work in the BabyBear field $\mathbb{F}_p$, $p = 15 \cdot 2^{27} + 1 \approx 2^{31}$. Integers map into $\mathbb{F}_p$ without wrap-around as long as they are small, and two checks ensure this. The verifier range-checks every claim ($|z| < 2^{29}$), and setup checks that every honest pre-activation is in the same range: $128 \sum_c |w_{ic}| + |b_i| < 2^{29}$ for every row i of a product, whose inputs are int8 and so at most 128 in magnitude (for an embedding, $\max_c |w_{ic}| + |b_i|$). A wrong claim and the correct value then differ by less than $2^{30} < p$, so they remain distinct in $\mathbb{F}_p$.

Checking (S3): committed weights. In K and K_{pre} the verifier holds the weights and computes u_ℓ itself, so (S3) holds by construction. In setting C it cannot, so the prover sends u_ℓ , and the verifier must check that $u_\ell = \chi^\top A_\ell$ for the committed A_ℓ . Opening a few entries of u_ℓ would miss one changed entry, as a path misses one neuron, so we commit to an *encoded* copy of the weights, as in the Ligerio and Brakedown commitments [1, 18]. At setup every row of A_ℓ is Reed–Solomon encoded [30]: the polynomial whose coefficients are the row is evaluated at $n_\ell = 4\bar{k}$ points, where $\bar{k} \geq k$ is a power of two (Section 3.6 relaxes this), giving $\hat{A}_\ell$. The commitment C_M holds one Merkle root [27] per weight matrix, over the columns of $\hat{A}_\ell$. Encoding is linear, so $\chi^\top \hat{A}_\ell = \text{Enc}(\chi^\top A_\ell)$. The verifier encodes the claimed u_ℓ itself, opens t random columns c with their Merkle paths, and checks $\chi^\top \hat{A}_{\ell,c} = \text{Enc}(u_\ell)_c$. If u_ℓ is wrong, $\text{Enc}(u_\ell)$ and $\text{Enc}(\chi^\top A_\ell)$ are distinct codewords whose difference is a nonzero polynomial of degree below k , so they agree on at most $k-1$ of the n_ℓ positions, and t distinct columns all miss the difference with probability at most $((k-1)/n_\ell)^t < 4^{-t}$. Because C_M is computed honestly, every committed row is a codeword and no proximity test is needed.

3.5 Protocol and soundness

Setup happens once per model: the owner encodes and commits the weights and publishes C_M . Figure 2 shows one query of the basic protocol, with r vectors χ and t columns per layer; besides recomputing the operations without weights (S1) and the checks of (S2) and (S3), the verifier range-checks the claims.

THEOREM 3.2 (SOUNDNESS). *If C_M is honest and the prover cannot find a hash collision, then for any claimed $y \neq M(x)$ the verifier accepts with probability at most $\varepsilon = \sum_\ell [p^{-r} + (\frac{k_\ell-1}{n_\ell})^t]$.*

Proof. Some claim is wrong, since y is one; let ℓ be the first layer with a wrong claim. Its input X_ℓ is correct, because the verifier computed it from x and correct earlier claims (S1), and the range check keeps the wrong claim distinct from the true value in $\mathbb{F}_p$. So $Z_\ell \neq A_\ell[X_\ell; 1]$ for the committed A_ℓ , and the prover passes only by fooling a random check: with the true u_ℓ , Eq. (1) fails except with probability p^{-r} (S2); with another u_ℓ , a column check fails except with probability $((k_\ell-1)/n_\ell)^t$ (S3), since the opened columns are the committed ones unless the prover finds a hash collision. A union bound over the layers finishes the proof. $\square$

Parameters. For a target λ and L weight layers we set $\beta = \lambda + \log_2(2L)$, $r = \lceil \beta / \log_2 p \rceil$ and $t = \lceil \beta / 2 \rceil$; since $n_\ell \geq 4k_\ell$, each of the $2L$ terms of ε is at most $2^{-\beta}$, so $\varepsilon \leq 2^{-\lambda}$ (for VGG-16 at $\lambda=128$, $r=5$ and $t=67$). Interactively, each challenge is expanded with SHAKE-256 from a fresh 256-bit random key; deriving the keys from a hash of the transcript (Fiat–Shamir [13]) makes the protocol non-interactive. It is round-by-round sound [5] (once the claims are fixed only the first wrong layer’s χ can make them pass, and once u_ℓ is fixed only that layer’s columns can), so a prover that makes Q random-oracle queries succeeds with probability at most $Q 2^{-\beta}$ plus that of finding a collision [20], which 64 extra bits in β keep below $2^{-\lambda}$ for $Q \leq 2^{64}$.

Known weights. In K the verifier computes u_ℓ itself and messages (3)–(6) are omitted. In K_{pre} it computes the secret χ and u_ℓ once per model, so the proof is the claims alone. The prover learns each verdict and could repeat an accepted forgery, but until one is accepted the verdicts do not depend on χ , so a false acceptance occurs among Q queries that reuse χ with probability at most $Q \sum_\ell p^{-r}$.

3.6 Costs and optimisations

Costs. The prover runs the forward pass, computes u_ℓ and opens t columns, which it recomputes from W_ℓ ; its extra work, $O((r+t) \sum_\ell N_\ell k_\ell)$, does not depend on the number of pixels or tokens. With 4-byte claims and field elements and 32-byte hashes, a proof in setting C has $|\pi| = 4 \sum_\ell N_\ell t_\ell + 4r \sum_\ell k_\ell + 4t \sum_\ell N_\ell + 32 \sum_\ell h_\ell$ bytes: the claims, which grow with the input, and the folded rows, the t opened columns of N_ℓ entries and the $h_\ell \leq t \log_2 n_\ell$ hashes of each layer’s Merkle multiproof, which do not. For Llama-2-7B at 64 tokens these are 349, 23, 385 and 4 MB of the basic protocol’s 762 MB proof. In K and K_{pre} the proof is the claims alone.

Optimisations. The parameters above keep the soundness argument short but leave the proof larger than it needs to be. We call the protocol so far the *basic* protocol, and the one with the four changes below and the engineered verifier described after them the

optimised protocol, which Section 4 measures and Section 4.5 compares with the basic one. The changes alter only how the weights are committed and checked, which positions of the last block are proved for a next-token output, and how messages are sent, and Theorem 3.2 holds in the form

$$\varepsilon \leq \sum_j \left[p^{-r} + \prod_{i < t_j} \frac{m_j - 1 - i}{n_j - i} \right] \quad (2)$$

over the encoded matrices j (a looked-up embedding table adds no term), with message length m_j , codeword length n_j and t_j distinct opened columns. This product bounds the chance that all t_j positions fall among the at most $m_j - 1$ where two distinct codewords agree. (i) *Columns and codes*. Each Merkle tree opens the fewest distinct columns that keep the term of each of its matrices below $2^{-\beta}$, and matrices with codewords of equal length and similar column counts share one tree and one multiproof. A public *planning rule* picks, among six candidate rules, the codeword lengths and layouts that send the fewest bytes per query besides the claims, using at most twice as many encoded entries at setup as the basic protocol, or 2^{34} if that is more. (ii) *Transposed layout*. Committing the encoded rows $\hat{A}'$ of $A^\top$ lets the verifier combine positions instead of neurons: it draws r vectors $\chi' \in \mathbb{F}_p^T$ (the T unit vectors when $T \leq r$), computes $z = Z\chi'$ and $w = [X; 1]\chi'$ itself for each, and checks $\text{Enc}(z)_c = (w^\top \hat{A}')_c$ on t opened columns of k entries. Since $w^\top \hat{A}' = \text{Enc}(Aw)$, a wrong claim passes only if all r products $(Z - A[X; 1])\chi'$ vanish or all t columns fall where two distinct codewords agree (m_j in Eq. (2) counts the rows it stacks). No folded row is sent and the opening shrinks from $rk + tN$ to tk field elements, which pays off for wide layers such as GPT-2’s output layer; projections of one input may share one such matrix. (iii) *Decoders*. Embedding tables without a bias are committed as Merkle trees over their rows, as in CommitLLM [6], so a wrong row needs a hash collision. For a next-token output, both parties compute the last block’s keys and values at every position and the rest of that block at the last position only, which gives the same logits. (iv) *Compact encoding*. Claims travel at 16–19 bits instead of 32 and field elements at 31; the verifier decodes before checking, and Fiat–Shamir absorbs the encoded bytes. The resulting plan (lengths, layouts, groups) follows from public shapes and is part of C_M . In K and K_{pre} , which open no columns, the optimised protocol keeps (iii), with the verifier reading the embedding rows itself, and (iv). Separately, *verifier engineering*, which changes no proof byte, makes the verifier evaluate $\text{Enc}(u_t)$ only at the opened positions, batch its exact products, compute attention in int32 (as the released prover also does) and stream the claims to a GPU.

4 Results

[Red: basic-protocol placeholder (red –: no basic run) for an optimised time still to be (re-)measured.]

4.1 Experimental setup

Models. LeNet-5 (MNIST) and VGG-11, VGG-16 and ResNet-18 (CIFAR-10) are the CNN architectures that zkCNN, ZKML and Bionetta report on; Table 3 compares only the published versions that match ours (ZKML’s ResNet-18, with 281K parameters, does

not). A 224×224 dog-vs-cat ResNet-18 trained on ImageNet classes stands in for the unreleased model of [2], and a dog-vs-squirrel one for its substitute $\tilde{M}$; the MNIST MLP (784–512–256–10) is the network attacked in Section 4.2. We trained all of them, and int8 quantisation changed accuracy by at most 0.4 points. For language models we use the shapes of GPT-2, OPT-125M to OPT-13B, Llama-2-7B/13B and Qwen3-4B with random int8 weights. The costs depend only on the shapes, except the size of the compact encoding of Section 3.6, which depends on the values; we measure it on the same random weights, so it may differ for real checkpoints. Every decoder up to 13B is built in full, at up to 2,048 tokens.

Real weights. To check that the integer graph is a usable language model, we load the real OPT-125M, OPT-1.3B and OPT-6.7B checkpoints into the same graph, with one extra bias on the LM head, left over from folding the final LayerNorm into it. On 4,064 tokens of WikiText-2 their perplexity is 73.8, 37.4 and 36.0, against 64.7, 34.9 and 27.1 in fp32. These values depend on two public constants, the gain G of the integer normalisation (4, 3 and 2) and the activation-calibration percentile, which we chose on the evaluation text, so these perplexities are slightly optimistic.

Hardware and baselines. The prover is one NVIDIA L40S (48 GB) and the verifier 8 threads of an AMD EPYC 9334; times are medians over a run’s queries. Of the 30–70B shapes only 1- and 2-block builds fit. From them we extrapolate proof sizes but not times; where a full build exists, the extrapolated size is within 2.83% of it. Published costs are taken as reported on their own hardware, each checked against its source; for the path protocol we compute exact detection probabilities and bytes on our graphs and take its time per path from Anchuri et al. The basic protocol was measured at commit 9401431 (a second run on an RTX 2080 Ti reproduced all 1,978 hardware-independent values) and the optimised one with the released code (commit c8be5eb, and two successors that change only how long proofs are encoded and held in host memory, not a byte), which reproduces the basic proofs byte for byte.

4.2 Breaking the path protocol

Reproduction. Our implementation behaves as Anchuri et al. describe. On MNIST versions of four of their substitute-model settings, honest provers were accepted in all 3,000 runs of each, and substitute models were caught in 99.8–100% of runs, even an 8-bit quantised copy that agrees with M on all 150 evaluated queries. Without the prescribed tolerance of 10^{-4} , an honest prover is accepted in 21–37% of runs, because a layer-wide matrix product and a single-neuron dot product round differently.

Attacks on MNIST. We compute each attack’s exact probability of passing one path by dynamic programming over the paths. The forgeries of [2] (its substitute model, Appendices E and F and Section 7.2) corrupt hundreds of the MLP’s 778 trace nodes and are always caught, while ours leave 1–14 inconsistent nodes. One tampered neuron is accepted with the predicted probability $1 - 1/N$ (99.6–99.8%). Triggered by a 3×3 bright patch, the backdoor returns the honest output on every clean query and succeeds on every triggered query not already in the target class. It passed 99.96% of 2,300 end-to-end runs, within sampling error of the predicted

99.8%. Keeping each tampered value below its neuron’s 95th percentile takes about five neurons and still passes 99% of the time under uniform sampling. With 250 paths, the single-neuron attack is accepted 61% of the time.

Other samplers. Against an adversary who knows the rule, uniform sampling detects 0.20% of attacks, a static importance score (outgoing weight magnitude) 0.11% and gradient saliency 0.016%. The contribution-weighted sampler detects nothing: zeroing about 14 of the 512 neurons, their most common value (65% of inputs), flips the prediction, and end-to-end runs accepted 2,000 of 2,000 attacks. A floor $\delta = 1$ raises detection to 2% (10 $\times$ uniform); other floors give 0.9–6 $\times$.

Larger models. On every CNN and every test query, changing one penultimate neuron suffices to flip the prediction. One path catches it with probability 1/84 (LeNet-5) to 1/512 (VGG, ResNet), and a change to the least-visited neuron with probability as low as 1/28 million. This includes the dog-vs-cat ResNet-18, a binary classifier, for which Anchuri et al. argue that their analysis extends to full soundness because the adversary can only flip the output. Reaching 2^{-40} takes 2,316–14,182 paths, which open the whole model and trace (Figure 3). On Llama-2-7B one path detects the attack with probability 1/11,008, and 2^{-40} needs about 305,000 paths (≈ 13.7 GB with the fp16 weights of [2], at 64 tokens).

A real LLM. On the real OPT-6.7B (fp16), the attacker ranks a few FFN neurons of layers 8, 16, 24 and 31 at the last position, by gradient and by their direct effect on the logits. It then finds the smallest change to one of them that changes the next token. One neuron sufficed for all 40 prompts. A fixed-target backdoor is more constrained: because of the final LayerNorm, a last-layer neuron can force only its *saturation token*, so the 16,384 neurons reach 6,187 of the 50,272 tokens. The backdoor succeeds for “back” (40/40 triggered prompts, clean prompts unchanged) and fails for “hacked” (0/40). Its values exceed the neurons’ natural range, which the path protocol does not check; one path detects it with probability 1/16,384, and 2^{-40} needs 454,248 paths.

4.3 Soundness in practice

In the optimised runs the verifier accepted all 3,792 honest queries and rejected all 2,512 attacks. The 1,848 image-model attacks, at $\lambda=40$ with committed weights, were one changed value, one changed penultimate neuron, one changed logit, 1% of the weights perturbed, the substitute $\hat{M}$, and two forgeries that each pass every check but one. Every language-model run added +1 to one middle-layer pre-activation (106 attacks, 20 of them on 1- and 2-block builds of the 30–70B shapes), and for five models from OPT-1.3B to Llama-2-13B we also ran the image-model attacks that apply to them. Freivalds’ check rejected 1,601 attacks and the column check against the code 853, including every forged folded row and every tampered claim of a layer in the transposed layout; the Merkle check rejected all 56 forged columns, so each of the three checks is needed (a range check caught the remaining two). The basic protocol’s run accepted all 8,709 honest queries and rejected all 2,895 attacks, 804 of them on language models.

Figure 3 plots security against the bytes sent per query for the image models, each with its own marker, orange for its paths and

Optimised at 128 bits sends less than paths at 40 bits

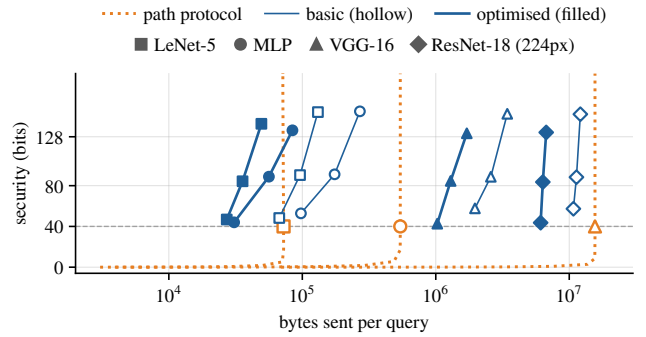


Figure 3: Security against bytes sent: our proofs buy far more security per byte than the paths

Table 1: Image models: proving takes milliseconds

Model	Params	acc.	committed (C)			K_{pre}	
			prove	verify	proof	verify	proof
MLP (MNIST)	536K	97.7%	3.8 ms	1.7 ms	84.3 kB	0.7 ms	1.9 kB
LeNet-5 (MNIST)	62K	99.3%	5.3 ms	2.5 ms	49.4 kB	1.6 ms	14.2 kB
VGG-11 (CIFAR)	9.8M	91.1%	12.8 ms	11.7 ms	1.0 MB	6.5 ms	334 kB
VGG-16 (CIFAR)	15.2M	93.0%	17.7 ms	17.2 ms	1.7 MB	10.1 ms	610 kB
ResNet-18 (CIFAR)	11.2M	94.2%	20.0 ms	24.9 ms	2.3 MB	16.4 ms	1.3 MB
ResNet-18 (224px) [†]	11.2M	95.0%	22.9 ms	64.8 ms	6.7 MB	59.2 ms	5.7 MB

[†]Our ImageNet-trained counterpart of the classifier of [2].

blue for our protocols. Security, in bits, is $-\log_2$ of the probability that a wrong output is accepted: for the paths, that the single-neuron attack is missed, and for ours, the proven bound at $\lambda=40, 80$ and 128. The paths reach 40 bits only by opening the whole model and trace (the vertical lines). The optimised protocol lands close to its target (43–47 bits at $\lambda=40$ and 131–141 at $\lambda=128$) and, on every model, reaches 2^{-128} with fewer bytes than the paths need for 2^{-40} . The basic protocol overshoots its target because r and t are rounded up (48–58 and 150–153 bits), and at $\lambda=40$ it sends 1.1–8.3 $\times$ fewer bytes than the paths need for 2^{-40} .

4.4 Cost

Image models. Table 1 gives the cost of one query to the optimised protocol at $\lambda=128$ (acc.: the int8 model’s accuracy). With committed weights the prover needs 3.8–22.9 ms per query and the verifier 1.7–64.8 ms. The proof, 49.4 kB–6.7 MB, is 1.3–9.3 $\times$ smaller than the int8 weights for the MNIST and CIFAR models and 1.7 $\times$ smaller for the 224-pixel ResNet.

Language models. Table 2 proves the next-token logits of one prompt at $\lambda=128$ with random int8 weights; its GPU column runs our verifier on the GPU. With a 64-token prompt GPT-2 is proved in 76.1 ms and Llama-2-7B in 1.8 s, and Llama-2-7B’s 325 MB proof is about 21 $\times$ smaller than its int8 weights. At 2,048 tokens Llama-2-7B is proved in 17.6 s, but the proof grows to 6.6 GB and CPU verification to 2.8 min. The proof grows because the claims grow with the prompt (the K_{pre} proof, which holds only claims, is almost as large), and the verifier because it recomputes attention over all

Table 2: Language models: seconds to prove, but proofs grow with the prompt

Model	Params	committed weights (C)				K _{pre}	
		prove	verify	GPU	proof	verify	proof
64-token prompts							
GPT-2	124M	76.1 ms	76.9 ms	79.9 ms	14.8 MB	61.8 ms	10.9 MB
Qwen3-4B	4.0B	1.2 s	712 ms	—	223 MB	692 ms	160 MB
Llama-2-7B	6.7B	1.8 s	1.0 s	431 ms	325 MB	698 ms	196 MB
Llama-2-13B	13.0B	2.7 s	1.5 s	—	469 MB	1.1 s	313 MB
2,048-token prompts							
OPT-125M	125M	396 ms	4.8 s	318 ms	348 MB	4.6 s	341 MB
OPT-1.3B	1.3B	3.8 s	27.4 s	2.6 s	2.1 GB	27.5 s	2.0 GB
OPT-6.7B	6.7B	7.7 s	46.5 s	6.4 s	5.5 GB	50.1 s	5.4 GB
Llama-2-7B	6.7B	17.6 s	2.8 min	6.9 s	6.6 GB	58.2 s	6.4 GB
Llama-2-13B	13.0B	17.6 s	3.7 min	—	10.5 GB	3.7 min	10.3 GB

token pairs; in setting C a GPU verifier is 11–25× faster there and 1.0–2.4× at 64 tokens. Prompts whose claims are all sent before the challenge can share one set of challenges and columns; in batches of eight 64-token prompts, Llama-2-7B’s prover then needs 0.39 s and its proof 219 MB per prompt. For Llama-2-70B the extrapolated proof at 64 tokens is 1.7 GB, 41× smaller than its int8 weights.

Figure 4 places these runs among the published ones, with our verifier on 8 CPU threads and the published systems (and the path of [2]) as reported on their own hardware. Part of the prover gap is our GPU, since all the zkSNARKs except zkLLM ran on CPUs.

4.5 Effect of the optimisations

We compared the two protocols on the same machines, in setting C, interactive and at $\lambda=128$, on LeNet-5, VGG-16 and ResNet-18 (CIFAR), GPT-2 with 64- and 512-token prompts and Llama-2-7B with 1- and 64-token prompts; for the language models the optimised runs also compute the last block at the last position only. On average (geometric mean) the optimised protocol sends 61% fewer bytes (2.0–4.2× fewer), verifies in 77% less time (2.3–10.9× faster) and proves in 28% less time. Much of the verifier gain comes from verifier engineering, which also speeds up the basic proofs (468 to 190 ms for GPT-2 at 64 tokens). The prover gets faster too, except on the two MNIST models (4.3 to 5.3 ms for LeNet-5 and 3.1 to 3.8 ms for the MLP), where encoding the claims costs more time than the planning rule saves (3.0 and 1.9 ms without the compact encoding). **At 2,048 tokens the optimised protocol proves Llama-2-7B in 17.6 s, verifies it on the GPU in 6.9 s and sends 6.6 GB, against 17.6 s, 6.9 s and 11.6 GB for the basic protocol.** Without any change to the proof, the released code’s prover and the streaming verifier already take 5.4 s and 1.7 s on the basic proof format, and the prover then takes 1.9× the plain int8 forward pass (the K_{pre} prover). On Llama-2-13B the same streaming verifier takes 9.1 s, 5.4× Llama-2-7B’s for a 1.9× larger model, with host memory peaking at 63 GiB, for reasons we have not found. The optimisations make the one-time setup longer (114–116 s instead of 9–10 s for GPT-2, about 8.6 instead of 4.5 min for Llama-2-7B).

4.6 Comparison with published systems

In Table 3, which compares the optimised protocol with published systems on matched models, our prover wins in every row, while

Table 3: Ours against published systems: our prover always wins (bold ↓: ours lower; ↑: ours higher)

System	Model (tokens)	prover	verifier	proof
zkCNN ^a	LeNet-5	78×↓	2.1×↓	1.1×↓
Bionetta	LeNet-5	664×↓	3.6×↓	72×↑
zkCNN ^a	VGG-16	4,422×↓	2.8×↓	6.4×↑
ZKML	VGG-16	31,910×↓	2.2×↑	180×↑
zkGPT ^b	GPT-2 (64)	229×↓	3.6×↓	164×↑
DeepProve ^c	GPT-2 (64)	360×↓	14×↓	1.3×↓
DeepProve ^c	GPT-2 (512)	995×↓	3.8×↓	3.6×↑
ZKTorch	Llama-2-7B (1)	1,213×↓	301×↓	8.3×↑
Maverick ^d	Qwen3-4B (8)	2.9×↓	1.2×↑	1.8×↓
zkLLM	OPT-125M (2,048)	187×↓	1.1×↓	2,467×↑
zkLLM	OPT-1.3B (2,048)	58×↓	2.9×↑	13,979×↑
zkLLM	OPT-6.7B (2,048)	71×↓	3.1×↑	35,153×↑
zkLLM	Llama-2-7B (2,048)	35×↓	2.9×↑	35,902×↑
zkLLM	Llama-2-13B (2,048)	46×↓	—	55,899×↑
zkLLM ^e	Llama-2-7B (2,048)	48×↓	—	—

^aVerifier on one core (ours: 8 threads). ^bNo prompt length stated; ours: 64 tokens.

^cTransparent configuration, proving every position (HyperKZG: 9.4 MB). ^dOurs in K_{pre}, $\lambda=40$, one thread. ^ezkLLM’s code ran on our L40S (—: not reported).

the verifier and the proof win in some rows and lose in others. Our verifier runs on 8 CPU threads, except in the 2,048-token rows, where we use our GPU verifier, since zkLLM also runs on a GPU; the prover times and proof sizes there are those of Table 2, as the choice of verifier does not change the proof. For the language models ours prove the next-token logits. Most systems we compare with produce non-interactive proofs, so against them we make ours non-interactive with Fiat–Shamir (Section 3.5), which costs us extra columns; zkLLM is itself interactive, so against it we run ours interactively. We use committed weights except against Maverick, whose client knows the weights and checks with its own private randomness, as K_{pre} does.

Prover. Ours is faster than every zkSNARK prover on its matched model, by **35×** (zkLLM, **Llama-2-7B**) to 32,000× (ZKML, VGG-16). Part of this is hardware, so we also ran zkLLM’s public demo on our L40S. It proves one layer’s components separately and reports no verifier time, so we time one layer (median of three runs on each of two layers) and multiply by the number of layers; the demo lacks zkLLM’s cross-layer optimisation and the attention output projection. For Llama-2-7B at 2,048 tokens this gives about 844 s, **48×** our **17.6 s**. We do not use the demo for Llama-2-13B, whose time per layer grows out of proportion (2.5× the 7B figure). **Verifier and proof.** Our verifier is slower only than ZKML’s on VGG-16 (2.2×), zkLLM’s on OPT-1.3B at 2,048 tokens (2.9×) **and on the three larger models (2.9–3.1×)** and Maverick’s (1.2×). Maverick’s 124.5 ms is its 87.1 ms of matrix checks plus the 37.4 ms in which its client replays the non-linear operations, as our verifier does (the client’s whole online time is 260.2 ms); ours takes 151.6 ms, 41.1 ms of it decoding the compact encoding. Without the compact encoding our Qwen3-4B run beats Maverick on all three costs (117 against 354.5 ms, 111.0 against 124.5 ms and 35.19 against 36.08 MB); our prover, however, runs on the GPU and Maverick’s on 64 CPU threads. Maverick also draws a fresh challenge per query, whereas K_{pre} reuses a secret one. On LeNet-5 and on

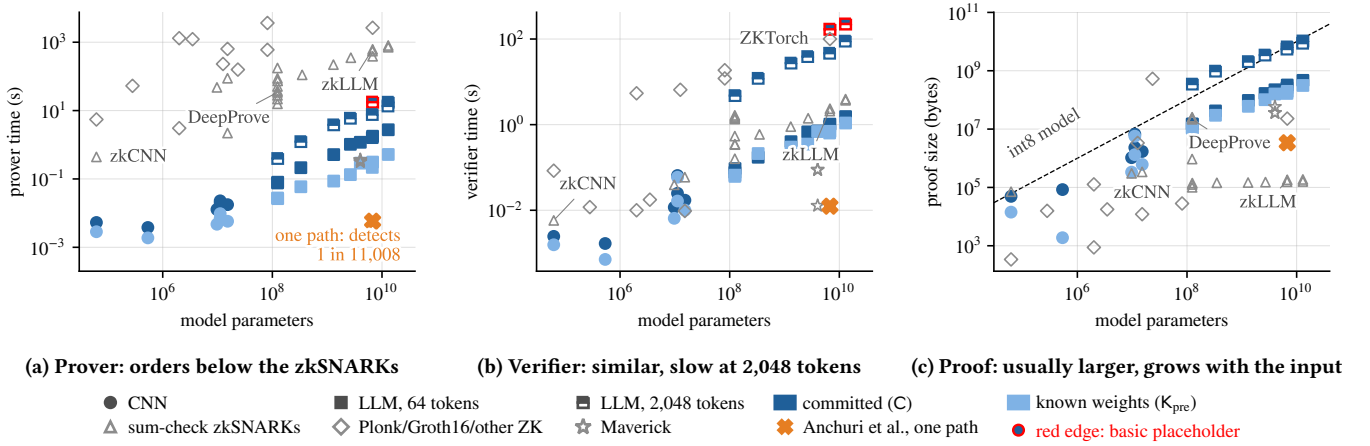


Figure 4: Cost against model size: ours (blue) proves far faster than published systems (grey), but usually sends more

GPT-2 with a 64-token prompt the non-interactive optimised protocol wins on all three costs: 5.7 ms, 2.8 ms and 63.3 kB against zkCNN’s 441 ms, 5.8 ms and 71.3 kB, and 95 ms, 97 ms and 16.6 MB against DeepProve’s 34.2 s, 1.35 s and 21.7 MB. zkCNN’s protocol has a zero-knowledge variant, which ours lacks, and DeepProve’s transparent proof certifies all 64 positions where ours certifies the next-token logits (Table 3, notes *a* and *c*). Elsewhere our proof is larger, by 3.6–180× against the other short-input proofs and by 2,500–56,000× against zkLLM’s at 2,048 tokens.

5 Discussion

Implications. The path protocol catches a provider who runs the wrong model, not one who edits the answer. Full soundness, however, does not need a zkSNARK prover: one random linear combination per weight layer gives a $2^{-\lambda}$ guarantee with no trusted setup and no assumption about how models behave. For MLPs and CNNs, which have no attention, the client verifies in milliseconds on a CPU and the proof is usually smaller than the model, so even a modest client can check every answer. For language models the verifier recomputes attention, so the protocol suits a gateway or an auditor with a GPU and a fast link rather than a phone; it works best for large models with short prompts, where the proof is far smaller than the model.

Limitations. (i) The proof grows with the activations; at 2,048 tokens it is mostly claims and gigabytes for 7B models, against zkLLM’s 157–183 kB. (ii) The verifier’s cost grows with the square of the prompt length: at 2,048 tokens the CPU verifier needs up to 3.7 min, and the GPU verifier is 2.9–3.1× slower than zkLLM’s. (iii) The provider must serve the integer model, which changed CNN accuracy by at most 0.4 points and raised the perplexity of real OPT checkpoints by 7–33%; the LLM costs were measured with random weights. (iv) Like the path protocol we assume an honest weight commitment, which the planning rule’s longer code-words make up to 13× slower to compute; a malformed one would need a proximity argument as in Ligerio. (v) The opened columns

reveal random linear combinations of each weight row, so the protocol is not zero-knowledge. (vi) We prove one forward pass, without a KV cache. (vii) The prototype is Python, and the same-GPU zkLLM comparison rests on its demo.

Future work. The claims, which the compact encoding roughly halves, could be replaced by a sum-check argument for the non-weight operations. Checking the attention products instead of recomputing them, with Freivalds’ check on claimed scores or a sum-check argument as in zkLLM [31], could make the verifier much faster at long prompts, at some cost in proof size. A native verifier and a KV cache would also help. Before any wider publication we will share the attack with the authors of [2].

6 Conclusion

A single tampered neuron defeats the path protocol of [2], on an MNIST network and on the real OPT-6.7B, and no sampling rule repairs it in the worst case. Checking every layer with Freivalds’ algorithm against a Reed–Solomon/Merkle commitment does: a wrong output is accepted with probability at most $2^{-\lambda}$, and the protocol rejected all 2,895 attacks on the basic protocol and all 2,512 on the optimised one. Its prover is one to four orders of magnitude faster than the zkSNARK provers, and on LeNet-5 and 64-token GPT-2 prompts it also verifies faster and sends less than zkCNN and DeepProve. Long prompts, where the proof is mostly claims, remain its weak point.

7 Contributions

[To be completed by the team.] Ofek Lavi: [...]. Erel Barzilay: [...]. Edo Kadosh: [...].

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